



Polymer electrolyte membrane water electrolysis: Restraining degradation in the presence of fluctuating power



Christoph Rakousky^a, Uwe Reimer^a, Klaus Wippermann^a, Susanne Kuhri^a,
Marcelo Carmo^{a,*}, Wiebke Lueke^a, Detlef Stolten^{a,b}

^a Forschungszentrum Jülich GmbH, Institute of Energy and Climate Research, IEK-3: Electrochemical Process Engineering, 52425 Jülich, Germany

^b Chair for Fuel Cells, RWTH Aachen University, Germany

HIGHLIGHTS

- Five identical single cells were used for constant and dynamic long-term testing.
- Time-based and hydrogen-based degradation rates are improved by current cycling.
- Intermittent operation of PEM water electrolysis is beneficial for durability.
- PEM water electrolysis is suited for operation with intermittent power sources.
- Degradation phenomena were analyzed and located.

ARTICLE INFO

Article history:

Received 30 May 2016

Received in revised form

23 November 2016

Accepted 30 November 2016

Available online 18 December 2016

Keywords:

Polymer membrane

Electrolyzer

Durability

Intermittent operation

Water electrolysis

ABSTRACT

Polymer electrolyte membrane (PEM) water electrolysis generates 'green' hydrogen when conducted with electricity from renewable - but fluctuating - sources like wind or solar photovoltaic. Unfortunately, the long-term stability of the electrolyzer performance is still not fully understood under these input power profiles. In this study, we contrast the degradation behavior of our PEM water electrolysis single cells that occurs under operation with constant and intermittent power and derive preferable operating states. For this purpose, five different current density profiles are used, of which two were constant and three dynamic. Cells operated at 1 A cm^{-2} show no degradation. However, degradation was observed for the remaining four profiles, all of which underwent periods of high current density (2 A cm^{-2}). Hereby, constant operation at 2 A cm^{-2} led to the highest degradation rate ($194 \mu\text{V h}^{-1}$). Degradation can be greatly reduced when the cells are operated with an intermittent profile. Current density switching has a positive effect on durability, as it causes reversible parts of degradation to recover and results in a substantially reduced degradation per mole of hydrogen produced. Two general degradation phenomena were identified, a decreased anode exchange current density and an increased contact resistance at the titanium porous transport layer (Ti-PTL).

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

Electricity generated from renewable sources such as wind and sunlight can help reduce the emission of CO_2 , but the power output of these technologies is innately subject to fluctuation. In order to generate hydrogen from these 'green' but intermittent sources by means of polymer electrolyte membrane (PEM) water electrolysis, the fluctuating profiles can either be directly used or, for instance,

capped at a maximum current density to be used as input power profiles. While it has been demonstrated that PEM water electrolysis is suitable for fluctuating input power with regard to its response time, not much attention has been paid to the effects of fluctuating profiles on long-term stability. However, an installed electrolyzer is intended to be operated for several thousand hours, and therefore discovering operating windows in which degradation is minimal is very important for implementing PEM water electrolysis on a large scale.

Degradation tests were carried out using either constant current modes [1–4] or highly volatile current densities, thereby reproducing a solar profile [5–8]. With Ir-based anode catalysts, the

* Corresponding author.

E-mail address: m.carmo@fz-juelich.de (M. Carmo).

lowest reported degradation rate is $0 \mu\text{V h}^{-1}$ at 1.6 A cm^{-2} / $\sim 1.9 \text{ V}$ [9]. A high degradation rate would be, for instance, $230 \mu\text{V h}^{-1}$ at 1 A cm^{-2} / $\sim 1.6 \text{ V}$ [2]. Other researchers report intermediate degradation rates [1,5,10,11]. However, comparing these results can be confusing, as they do not indicate intervals of current density or cell voltage that can be identified as operating windows with minimum degradation. This may be due to the fact that different research groups may choose different operating temperatures, test durations and cell setups. In a previous publication [12], we showed that the durability of PEM water electrolysis single cells is especially affected by the cell assembly. Therefore, to reach meaningful conclusions about the effect of different operating power profiles, they must be compared in a study with identical cells and setup.

Some degradation mechanisms have already been identified, which show, for instance, that the cationic contaminants of feed water cause a poisoning of the electrode layers and degradation of cell performance [10,11,13–15]. In one study, the authors reported that both the ohmic cell resistance and charge transfer resistance increase linearly over time, while low frequency capacitance decreases [5]. They used a power profile that comprises a variety of current densities between 0 and 2 A cm^{-2} at 80°C , and therefore it cannot be determined which current density interval was primarily responsible for the degradation. In another study on the effect of intermittent operation, stable cell performance was observed at 1 A cm^{-2} , with periodic current interruptions of 1 min [16]. However, it is still unclear, as to whether intermittent operation affects degradation at current densities $> 1 \text{ A cm}^{-2}$. In our previous publication [12], we demonstrated that the stability of the electric resistance of the anode Ti porous transport layer (Ti-PTL) greatly affects the durability of the cell at a high current density of 2 A cm^{-2} and that the use of platinum group metal (PGM)-based anti-corrosion coating significantly reduces cell degradation.

In this study, we compare degradation rates and report on the degradation phenomena we see with our cell setup under constant and fluctuating power profiles, using the economically important operating current densities of 1 and 2 A cm^{-2} at cell temperatures of 80°C . For this work in particular, we use pure Ti-PTL to identify operating windows in which the use of a PGM-based anti-corrosion coating can be avoided. These results can help to develop suitable ways to couple PEM water electrolysis to renewable sources.

2. Experimental

Five electrolysis cells were operated for 1009 h (about 6 weeks) with separate load profiles comprising constant and dynamic operation between 0 and 2 A cm^{-2} . In order to locate and understand the source of the degradation of cell voltage, electrochemical and physical characterization methods were employed.

2.1. CCM and cell setup

The catalyst coated membrane (CCM) samples were products of Greenerity GmbH. The anode catalyst layer contained IrO_2 and TiO_2 with an Ir-loading of 2.25 mg cm^{-2} . Pt/C was used as the cathode catalyst at a Pt loading of 0.8 mg cm^{-2} , while Nafion® 117 membrane was used as the electrolyte. All CCMs were equipped with a dynamic hydrogen reference electrode (DHE) [17–20] that was assembled in accordance with our previous publication [12]. CCMs were assembled in dry conditions. Two sheets of Toray paper (TGP-H 120) with a total thickness of $700 \mu\text{m}$ were used as the cathodic porous transport layer (PTL). Porous titanium sinter plates (GKN T3P, $42 \times 42 \times 1.3 \text{ mm}$) were used as anodic PTLs. Platinum-coated titanium bipolar plates with meander-type flowfields were used on both sides, with that on the cathodic side also gold-plated against hydrogen embrittlement. PTFE and Silicon gaskets sealed the cell

with an active cell area of 17.64 cm^2 . Five of these cells were then simultaneously tested.

2.2. Test bench and long-term testing

The cells were operated using a test bench developed in-house that allowed for the simultaneous operation of five single cells at individual load cycles. High purity feed water ($18.2 \text{ M}\Omega \text{ cm}$) was circulated through both the anode and cathode compartments (25 ml min^{-1}) in separate water circuits for each cell compartment. All ten water circuits were equipped with compartments containing ion exchange resin to maintain a high degree of water purity. The cell temperature of 80°C was reached by preheating the feed water to 75°C and incorporating additional heating cartridges to stabilize cell temperature at all current densities.

Before installation to the test bench, the assembled cells were purged with deionized water (80°C) for 4 h to remove potential ionic impurities. Following installation to the test bench, the cells were kept at 80°C under flowing feed water until the end of the durability test.

The different current density protocols for long-term testing are given in Table 1. They comprise constant (cells A and B) and intermittent current density operations (cells C, D and E). The durability tests were stopped after 1009 h (about 6 weeks). To facilitate the discussion, the test period is rounded down to 1000 h in this article. Due to its non linear voltage response, the test period of profile B was extended to a total duration of 1150 h. The results of this extension period are outlined in our previous publication [12]. Electrochemical impedance spectroscopy (EIS) and polarization curves (j-E curves) were recorded regularly. It should be noted that except for the time periods in which EIS was recorded, cells A and B were never operated at voltages of less than 1.4 V, the lowest voltage that occurred during polarization curves.

Half-cell potentials were measured using the DHE reference electrode according to our previous publication [12]. Only one DHE could be operated at a time. Therefore, the DHE currents ($6\text{--}12 \mu\text{A}$) had to be restarted for each measurement. The DHE voltages between the Pt wires ($1.7\text{--}1.9 \text{ V}$) increased upon DHE start-up and reached constant values after about 30 min. All reference electrode data were recorded after 60 min so as to gain comparable data.

2.3. Electrochemical characterization

Polarization (j-E) curves were plotted for the cells and EIS was performed. Specifically, polarization curves were recorded every 168 h with the DC power supply and run galvanostatically from 0.03 A cm^{-2} to 3 A cm^{-2} with a limiting cell voltage of 2.2 V and a 1 min hold per step. For EIS measurements, the cell current had to be interrupted in order to unplug the DC power supply and connect the cell to the potentiostat (Biologic HCP 1005). EIS spectra were recorded at current densities (and amplitudes) of 0.06 (0.006) - 0.11 (0.011) - 1 (0.06) - 1.5 (0.06) - 2.0 (0.06) A cm^{-2} in a frequency range of 100 kHz and 100 mHz. Each step was preceded by a 3 min conditioning period at the respective current density. The equivalent circuit $R_Q(R_{CT}Q_{DL})$ was fitted to the spectra using Zahner's

Table 1
Protocol for durability testing.

Mode	Type	$j/\text{A cm}^{-2}$	Interval
A	Const.	1	1009 h
B	Const.	2	1009 h
C	Dyn.	2–1	6 h–6 h
D	Dyn.	2–0	6 h–6 h
E	Dyn.	2–0	10 min - 10 min

THALES software. R_{Ω} represents the ohmic cell resistance, R_{ct} the combined charge transfer resistance of both hydrogen and oxygen evolution reactions and Q_{DL} a constant phase element to describe the double layer capacity of the porous electrodes.

2.4. Physical and chemical characterization

Electrical through-plane resistance of the Ti-PTL were measured with an in-house device at 0.06 A cm^{-2} , room temperature and a surface pressure of 346 N cm^{-2} . For this purpose, the sample was sandwiched between and contacted by two carbon nonwoven gas diffusion layers by Freudenberg FCCT KG. For cross-section imaging, a strip of the dried CCM was embedded in epoxy resin and polished. Images were recorded using a Carl Zeiss Scanning Electron Microscope (SEM) Gemini Ultra Plus. To determine the elemental distribution across the CCM, EDX linescans were performed across the thickness of the CCM with a SiLi detector from Oxford Instrument Pentax FET. Water samples were taken weekly and its conductivity was determined with a Mettler Toledo Seven Multi. Imaging of the catalyst samples was performed by means of an FEI Tecnai F20 transmission electron microscope (TEM) with an accelerating voltage of 200 kV. For this purpose, parts of the electrode layer were scratched off the aged CCM and a fresh CCM of the same lot, and were then ultrasonically dispersed in a mixture of isopropanol and water (50:50 vol-%) and coated on a copper grid (PLANO S160). X-ray diffraction (XRD) was used to determine the crystal size of the scratched off catalyst samples of the aged CCM, as well as a fresh one. XRD was performed using a STADIP transmission diffractometer with Cu-K α 1-radiation at a wavelength of $\lambda = 1.54056 \text{ \AA}$. Crystal sizes were calculated using Scherrer's equation via the SiroQuant[®] software, version 4.

3. Results and discussion

3.1. Cell durability

An increased cell voltage is a sign of degraded cell performance. As can be seen in Fig. 1, the performance degradation depends on the testing protocol. Both the current density and cycling frequency have a significant effect on the cell's durability.

Constant operation at 2 A cm^{-2} (cell B) resulted in the highest degree of degradation (increase in cell voltage). An analysis of the degradation phenomena of cell B was published in our previous work [12]. There, we attributed the irreversible part of the observed performance degradation to increasing ohmic cell resistance and decreasing anode exchange current density due to electrode poisoning by Ti cations. We also reported the positive effect of a current interruption, which improved cell voltage after restarting the electric current [12]. In particular, this effect can be seen in Fig. 1B at 500 h and can now be further investigated using the different current density profiles.

In contrast to the pronounced degradation in mode B, constant operation at the lower current density of 1 A cm^{-2} (mode A) results in a stable cell voltage over time. It can therefore be concluded that during these measurements no degradation occurs at cell voltages of 1.70 V or at 1 A cm^{-2} , but that pronounced degradation is seen at 2 A cm^{-2} , starting at voltages of 1.84 V.

Whether higher voltage or higher current density are more detrimental remains unclear; both parameters are reasonable. Higher cell voltages accelerate corrosion processes [21,22] and should lead to increased ohmic resistances due to passivating oxide layers or catalyst dissolution. However, stable cell performance at high cell voltages ($>2.0 \text{ V}$) was published over 60,000 h with minimal degradation [9], suggesting that depending on the cell design, high voltages may not be the only possible driver for

degradation. Higher current densities, on the other hand, may also cause degradation. They are accompanied by: i) increased gas bubble formation, which may result in increased mechanical impact on the electrode layers and potentially cause CCM-delamination; ii) higher oxygen concentration in the anode compartment, which could accelerate corrosion and passivation processes; iii) increased consumption of feed water, leading to the accelerated accumulation of potential contaminants in the CCM (as indicated by Ref. [23]); and iv) the potential formation of hotspots in the case of bad water management. If current and voltage were decoupled from one another, e.g., by varying membrane thickness or catalyst activity, more insight could be gained into solving this question. We intend to trigger a discussion as to which parameter is more detrimental with respect to the degradation. For the sake of clarity, in this manuscript we report the degradation with respect to the current density used, as this was the parameter we controlled.

The dynamic modes, C, D and E, show intermediate increases in cell voltage with respect to the constant modes A and B (Fig. 1). It must be noted that both the current density and switching times affect the degradation. Of the dynamically operated cells, the highest degradation is observed for cell C, which cycles between 1 and 2 A cm^{-2} . Further reducing the lower current density to 0 A cm^{-2} (modes D, E) leads to an even smaller voltage increase over time. In particular, the cycling interval plays an important role. Long cycling intervals of 6 h (mode D) result in less degradation than shorter cycling intervals of 10 min.

Consequently, during the 1000 h of testing, the amount of hydrogen produced by the five cells was different. Thus, two different degradation rates must be contended with. First, the frequently used average degradation rate, which is calculated from the voltage increase ΔE_{LT} and total operation time of 1000 h and termed δ_{chrono} . Second, to account for the different current densities and idle periods, we establish the 'molar voltage degradation rate', δ_{molar} , as a figure of merit to compare different operating profiles. This can be thought of as the average voltage increase per mole of hydrogen produced. Both degradation rates are compared in Table 2. The five current density profiles used rank similarly for both δ_{chrono} and δ_{molar} and only differ in terms of the position of the profiles C and E, whose values are close to each other. This agreement indicates that the different amounts of hydrogen produced do not explain the differences in observed degradation. From this it can be concluded that it is possible to control the performance degradation by controlling the current density profile. Low current density operation produces no degradation. If higher current densities are necessary, such as to reduce installation costs, a given mole of hydrogen produced by profile D is associated with 84% less cell performance degradation than if it was produced by the same cell operated with profile B. This can be summarized as follows:

- Constant operation: At 2 A cm^{-2} degradation is high, whereas no degradation occurs at 1 A cm^{-2} . Therefore, for the cell configuration discussed here, long and uninterrupted periods of operation at 2 A cm^{-2} should be avoided.
- Dynamic operation: Degradation rates at high current densities of 2 A cm^{-2} are reduced if repetitive periods of lower current density are used (mode C), preferably current interruptions (modes D, E).
- Current interruptions: They should be used with caution if they are applied to reducing the degradation rate. A smaller number of longer current interruptions are preferred to a higher number of shorter interruptions. Therefore, frequent shut-offs are not recommended.

Our results indicate that in terms of long-term stability, PEM electrolysis is well suited for operation with intermittent power

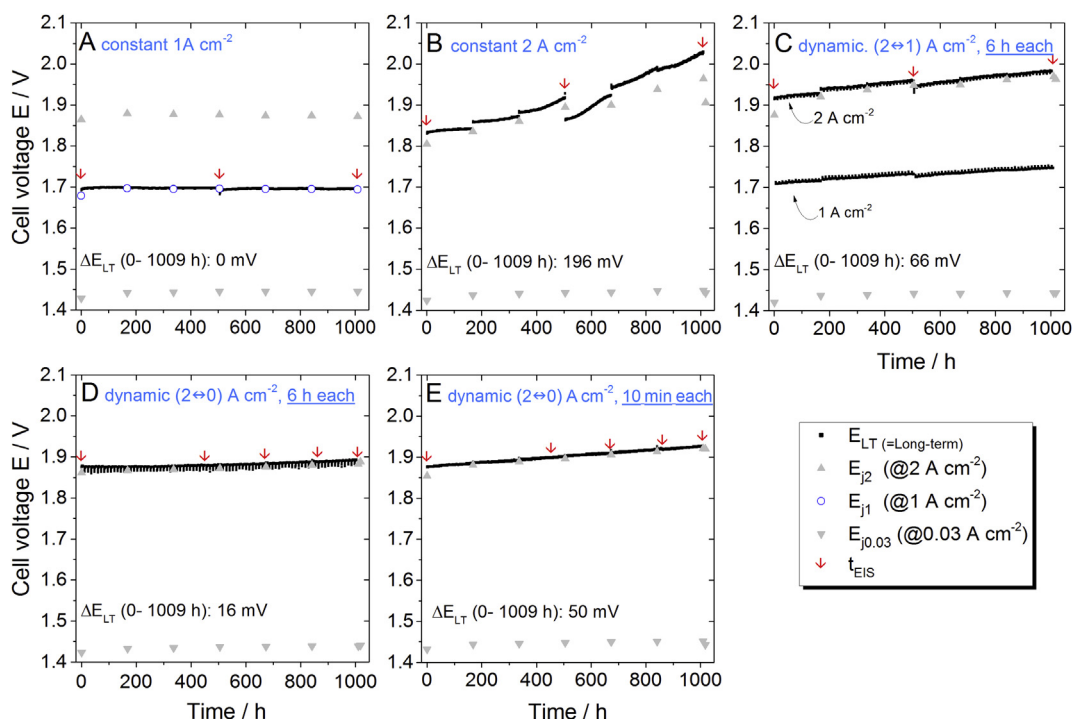


Fig. 1. Long-term development of the cell voltage for five different operation modes. A: 1 A cm⁻², B: 2 A cm⁻², C: alternating operation at 2 and 1 A cm⁻² for 6 h each, D: alternating operation at 2 and 0 A cm⁻² for 6 h each, E: alternating operation at 2 and 0 A cm⁻² for 10 min each. Voltages at 0.03, 1 and 2 A cm⁻² taken from the polarization curves as well as the moments when EIS were recorded are marked, too. The operating conditions significantly affect the performance durability. Dynamic operation is able to reduce the degradation, compared to constant operation at high current densities (B). Figure B: modified from Ref. [12] with permissions from Elsevier.

Table 2
Observed degradation rates in the durability test.

Mode	$\Delta E_{LT}/mV$	$\delta_{\text{chrono}}/\mu V h^{-1}$	$\delta_{\text{molar}}/mV mol^{-1}_{H_2}$
A	0	0	0
B	196	194	5.2
C	66	65	2.3
D	16	16	0.8
E	50	50	2.7

sources such as wind or solar. Current interruptions have a positive effect on the cell voltage after restarting the operation. It seems that reversible fractions of the degradation are reset when the current is interrupted, a phenomenon not yet fully understood [12]. The positive effect of a shut-off can clearly be seen in the long-term voltage trend in Fig. 1B and C at 500 h, where the cell voltages were interrupted in order to take EIS measurements. Upon restarting the current after EIS, the cell voltages had dropped by 54 mV (cell B) and 15 mV (cell C). Since the reversible property of the degradation builds up over time, a frequent current interruption can reduce the development of substantial reversible degradation, as they are frequently reset. Support for this hypothesis is provided by the fact that, because of the large number of regular current interruptions, cells D and E are unaffected by the additional current interruptions due to the EIS measurements.

3.2. Half-cell potentials

Fig. 2a) shows the cell voltage and both half-cell potentials. For all cells, the difference between the anode and cathode potential equals the cell voltage. Comparing the first data points of cells B-E, it is obvious that the anode potentials differ significantly more from each other (1.63–1.90 V) than the corresponding cell voltages (1.86–1.94 V). This variation between the cells can be explained by

the fact that the measured half-cell potentials depend on both the location of the reference electrode and the alignment of the electrode's edges [24–26]. As both factors depend on the cell assembly, they can differ slightly for each cell, but will not change with time. Therefore, the trends in half-cell potentials can be used to determine which cell compartments undergo degradation.

For all cells that exhibit degradation (cells B-E), the anode compartment shows higher degradation than that of the cathode. The constant cell voltage and constant half-cell potentials of cell A are in good agreement, indicating sufficient stability of the reference electrode potential during the test period. For cell B both anodic and cathodic overpotential increase. At 500 h, when the cell was stopped for EIS-measurements, the anode potential is constant, while both the cell voltage and cathode potential show a significant change. This indicates that the reversible degradation phenomenon stems from the cathode side. However, the performance improvement at 500 h for cell C is accompanied by a constant cathode potential and drop in the anode potential. Contrary to cell B, the reversible degradation for cell C seems to be located on the anode side. Thus, so far no location can be given for the effect of the reversible degradation.

To summarize, between 210 h and the end of the test, performance degradation can be attributed entirely (cells C-E) or mostly (73%, cell B) to the anode potential. This is measured between the anode bipolar plate and reference electrode. Hence, potentially degrading cell components include the anode catalyst layer, Ti-PTL, anode bipolar plate and all contact resistances in between.

3.3. Electrical resistance of Ti-PTL

Fig. 3 shows the electrical through-plane resistance, R_{PTL} , of the anode Ti-PTLs before and after the long-term test. Since temperature, current density and contact materials differ between this ex situ measurement and the in situ situation, values for R_{PTL} should

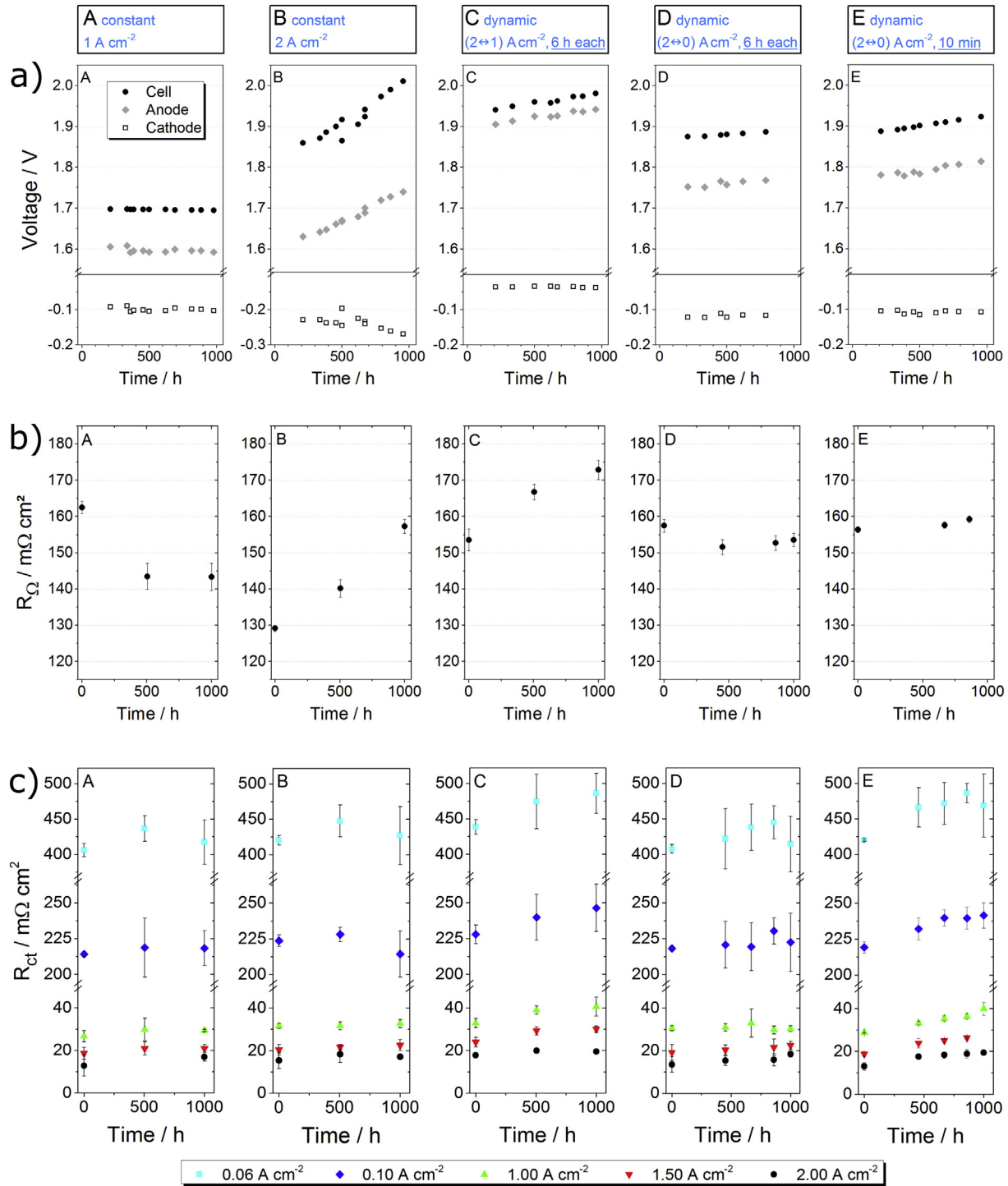


Fig. 2. a) Cell voltages and half-cell potentials for the highest current density of the respective profile (cell A: 1 A cm⁻², cells B-E: 2 A cm⁻²). Half-cell potentials were determined between the bipolar plates and DHE. While the cathode potentials are fairly stable, the anode potentials increase significantly. Cell degradation is therefore expected to take place predominantly on the anode side. Half-cell potentials could only be recorded after 210 h; b) Ohmic cell resistance from EIS measurements taken at 2 A cm⁻²; c) Charge transfer resistances R_{ct} obtained from fits to the EIS measurements, recorded at five current densities between 0.6 and 2.0 A cm⁻². Figures B modified from Ref. [12] with permissions from Elsevier.

only be evaluated qualitatively. The R_{PTL} of all PTLs increases over time, which may be due to the corrosion of titanium and subsequent thickening of the insulating surface oxide layer [27]. While R_{PTL} increases strongly for the Ti-PTLs B-E, it barely changes for Ti-PTL of cell A. This shows that operation of up to 1 A cm⁻² does not impact the functionality of the PTL, whereas higher current

densities (or the respective higher voltages) significantly degrade the performance of the PTL.

3.4. Electrochemical impedance spectroscopy (EIS)

EIS was carried out on all the cells. As an example for the

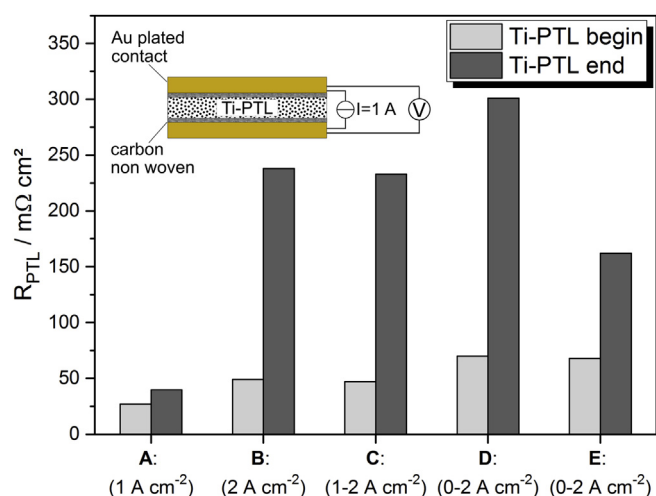


Fig. 3. Electrical through-plane resistance R_{PTL} of anodic porous transport layers (Ti-PTL) before and after the long-term test. The resistance remains unchanged when operated at 1 A cm⁻², but increased significantly for all PTLs that were operated with periods of 2 A cm⁻² (B–E).

appearance of the spectra, the ones for cell E are provided in the supplementary information (Fig. S1).

Ohmic cell resistances R_{Ω} and charge transfer resistances R_{ct} were evaluated and are displayed in Fig. 2b) and c). The trends in cell resistance are similar at all current densities; therefore, the exemplary R_{Ω} values at 2 A cm⁻² are shown in Fig. 2b). It can be seen that cells with a high increase in cell voltage (cells B, C and E) also exhibit a high increase in R_{Ω} . As these cells have been operated at an upper current density of 2 A cm⁻², it can be concluded that these operational modes are likely to provoke and consequently suffer from increased ohmic losses. In contrast, cells that only show a minor increase in cell voltage over time (cells A and D), also show a drop of R_{Ω} during the first 500 h. For cell A, this drop is highly pronounced and accounts for about 20 mΩ cm² until the 500 h mark is reached. Our previous publication [12] helps to illuminate why both an increase and decrease of R_{Ω} can be observed, even with the same type of CCM. We identified the pure Ti-PTL as a source of increasing resistance due to passivation. On the other hand, when an anti-corrosion coating is applied to the Ti-PTL, R_{Ω} decreases over time, even though the PTL-resistance remains constant. The drop was attributed to an adaptation of the CCM topology to the rough surface of the porous transport layers [12], leading to improved contact resistances between them.

Our data show that the use of PGM-based anti-corrosion coatings can be avoided for our setup when operated at a maximum of 1 A cm⁻² (section 3.3). The constant $R_{PTL,A}$ (Fig. 3) thereby allows the ohmic cell resistance R_{Ω} to decrease during the initial 500 h by 20 mΩ cm² (Fig. 2b), taking into account the above mentioned effect of CCM topology changes. At the operating current density of 1 A cm⁻², this drop should correspond to a decrease in cell voltage of 20 mV, which surprisingly is not observed during this period (Fig. 1A). The cell voltage actually observed is constant, indicating that the hypothetical performance improvement must be balanced by another mechanism, one that degrades the cell's performance and is not related to ohmic losses. Such a phenomenon can be a break-in of the catalyst, which is indeed observed for all cells in the form of an increasing charge transfer resistance in Fig. 2c). The trends in R_{Ω} of the remaining cells B–E can also be explained by two opposing trends. For these PTLs, increased electric resistances R_{PTL} were found and are superpositioned by the decreased contact resistance between the CCM and PTLs that arise from the topology

changes of the CCM over time. Both trends sum up to create increasing (cells B, C) or stable (cells D, E) values of R_{Ω} .

The interruption of the cell current seems to positively affect the ohmic losses. Cells B–E have been operated at upper current densities of 2 A cm⁻². Of these, the cells operated with current interruptions (D and E) show significantly less of an increase in R_{Ω} than cells that had never had a current interruption (B and C). However, the positive effect of the current interruption is difficult to locate. It cannot be investigated by means of EIS or any ex situ methods, as the reversible contributions recover upon stopping the cell current. For our setup, it can therefore be concluded that:

- High current densities of 2 A cm⁻² lead to an increase in the ohmic cell resistance R_{Ω} that can be attributed to increased R_{PTL} . However, at low current densities of 1 A cm⁻², R_{PTL} remains constant and R_{Ω} therefore decreases.
- If high current densities are to be used, current interruptions reduce the increase in R_{Ω} . The reason for this remains to be discovered.

The charge transfer resistances R_{ct} are presented in Fig. 2c) for all five current densities. Cells C and E clearly show a continuous increase in R_{ct} during the test period for all current densities. This increase is most pronounced for cell E, which experienced the largest number of current switching incidents in the test. During the first 500 h, all other cells show increasing R_{ct} as well. However, due to the higher uncertainty of the 1009 h fits, it cannot be said whether this trend continues until the end of the test for cells A, B and D. An increased R_{ct} may be an indication of a decreased exchange current density due to electrode degradation. To gain further insight into the reasons for the degradation we carried out the analysis of the polarization curves presented for cell B in our previous publication [12].

3.5. Analysis of polarization curves

For each cell, seven polarization curves were obtained during the long-term operation, as shown in Fig. 1. These curves are analyzed on the basis of a polarization curve model. The model and fitting procedure have been described in our previous work [12] and the main features are summarized in the following.

The cell voltage is calculated based on the Nernst voltage. Kinetic and ohmic losses are added according to a standard approach [28] (Eq. (1)). The kinetic losses at the anode and cathode are described by the Tafel Equation (Eq. (2)) through the parameter η_{act} , using the universal gas constant R , temperature T , charge transfer coefficient α and Faraday constant F . In this approach, both electrodes are combined into one effective electrode.

$$E_{cell} = E_{Nernst} + \eta_{act} + jR_{total} \quad (1)$$

$$\eta_{act} = \frac{RT}{\alpha F} \ln \frac{j}{j_0} \quad (2)$$

With Eq. (2), Eq. (1) contains two parameters for describing degradation effects. The exchange current density j_0 is a measured for electrode activity and is influenced, for instance, by the electrochemically active surface area of the catalyst. A decrease in j_0 indicates an overall loss of electrode performance without differentiation between anode and cathode. The parameter R_{total} reflects the cell's total ohmic resistance as fitted from the polarization curves. As is shown in Fig. 3 a change of R_{total} can be attributed to the contact resistance of the Ti-PTL. Table 3 shows the fitted parameters at the beginning of the experiment for the five different cells based on the values of $E_{Nernst} = 0.974$ V and $T = 353$ K (80 °C). It

Table 3Fitted model parameters for each cell at $t = 0$ h.

Cell	α	$j_0/\text{A cm}^{-2}$	$R_{\text{total}}/\Omega \text{ cm}^2$
A	1.16	$1.2 \cdot 10^{-09}$	0.166
B	1.40	$3.5 \cdot 10^{-11}$	0.147
C	1.17	$1.5 \cdot 10^{-09}$	0.180
D	1.19	$9.2 \cdot 10^{-10}$	0.170
E	1.25	$2.5 \cdot 10^{-10}$	0.164

can be recognized that the values differ significantly from cell to cell. Therefore, the cells cannot be compared directly, although the relative changes are discussed in the following.

Fig. 4a) shows the relative decrease of parameter j_0 . All cells seem to lose roughly 50% of their effective electrochemical surface area within the first 400 h. The reader is reminded that cells D and E, for example, were effectively operated for half the time compared to cells B and C. The actual load profile does not seem to have a significant influence on j_0 . Furthermore, the half-cell potentials in Fig. 2a) suggest that the main degradation effect relates to the anode, as was discussed in section 3.2.

Fig. 4b) shows the results for the parameter R_{total} . It increases over time. In particular, cells with a higher increase in R_{total} also show a higher voltage increase over time, indicating that R_{total} is an important degradation parameter. As in the case of R_{Ω} in section 3.4, two different groups can be identified.

Cells A and D show minor changes. For cell D, the resistance remains almost constant and for cell A, the value of R_{total} decreases slightly, to 98% of its starting value. It could be argued that the long breaks of 6 h for cell D led to an almost full recovery.

In contrast, for cells B, C and E, resistance increases significantly over time. The highest degradation can be observed for cell B which

was continuously operated at 2 A cm^{-2} (above 1.8 V). The trends in R_{total} for cells C and E are very similar, indicating that a decrease of cell voltage down to 1.7 V for 6 h (cell C) leads to a similar recovery effect as complete current interruptions for 10 min (cell E). It is as yet unclear whether current density or cell voltage is the main reason for degradation (see corresponding discussion in section 3.1). Cells B and C were operated at up to 2 A cm^{-2} without current interruptions and show the highest increase in R_{total} . The current was interrupted once at 505 h in order to measure EIS spectra (see Fig. 1). This interruption leads to a partial recovery of R_{total} , while the rates of change of R_{total} before and after this event seem to be roughly the same.

Table 4 compares the degradation rates at 2 A cm^{-2} of the recorded polarization curves ($\delta_{j-E, \text{exp}}$) and their modeling ($\delta_{j-E, \text{mod}}$). The model shows good agreement for such a simple approach. Through the two independent model parameters, the overall degradation can be divided into contributions of catalyst degradation and increases in total resistance. For cells B, C and E the total ohmic resistance clearly dominates the overall degradation, as can be seen in the last two columns in Table 4. In the case of cell D, a recovery process takes place during the 6 h breaks and therefore resistance accounts for only 33% of the total degradation effect. Again, it can be seen that the shorter but more frequent breaks of cell E seem to be less effective in reducing the increase in R_{total} . For cell A, the decrease of j_0 is almost compensated for by the decrease in total resistance, proving that the constant cell voltage is indeed a superposition of two opposing trends that are completed within the first 168 h. The decrease in R_{total} leads to improved cell voltage, so that the contribution of the catalyst to the overall degradation nominally exceeds 100% in Table 4. The decreasing R_{total} can be attributed to an adaptation of the CCM to the PTLs over the first hundreds of hours, as was discussed in section 3.4 and in our previous publication [12].

As a very rough estimate for all cells in this study it can be said that a 10% loss of j_0 , potentially by a reduction in catalyst surface area, leads to an increase of $3 \mu\text{V h}^{-1}$ (independent of the current density). An increase of 10% in total resistance accounts for $15 \mu\text{V h}^{-1}$ at 1 A cm^{-2} and $30 \mu\text{V h}^{-1}$ at 2 A cm^{-2} .

The decrease in exchange current density is consistent with the constant or slight increase of the charge transfer resistance. A changed exchange current density merely shifts the polarization curve towards higher cell voltages, but does not impact its slope within a given current density interval. Galvanostatic EIS therefore detects constant values for R_{ct} .

3.6. Locating the origins of the changed j_0

In order to locate the origins of the decreased exchange current density j_0 , feed water quality was analyzed and the electrode layers were characterized using XRD, EDX and TEM.

3.6.1. Water purity

The conductivity of the feed water was below $1.4 \mu\text{S cm}^{-1}$ for all water circuits throughout the entire test period. This shows that the test bench, along with the ion exchange resin, maintain high water purity and that ionic contamination of the electrodes from external ion sources in this testing setup is unlikely.

3.6.2. Analyses of the CCM

In our previous publication regarding cell B [12], we found Ti poisoning in the cathode catalyst layer. In brief, the Ti found in the cathode must be an oxidation product of anodic Ti components, as the bipolar plates were found to be without defect after the test and all remaining sources of Ti in the system are anodic. We therefore suggested that Ti ions either enter the anode from the Ti-PTL or

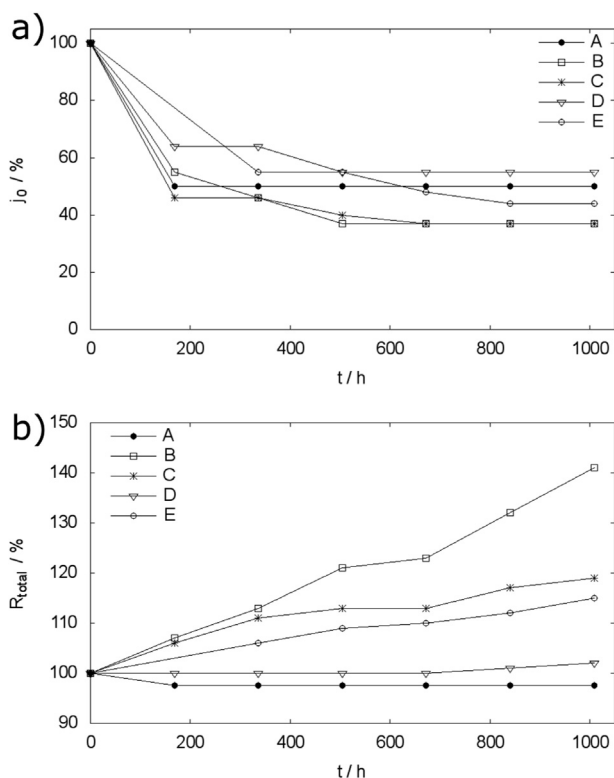


Fig. 4. a) Fitted values for j_0 normalized by $j_0(t = 0 \text{ h})$ (lines to guide the eye); b) Fitted values for R_{total} normalized by $R_{\text{total}}(t = 0 \text{ h})$ (lines to guide the eye). Graphs B: from Ref. [12] with permissions from Elsevier.

Table 4

Degradation at 2 A cm^{-2} as observed in experiment and model; ^{a)} for cell A, R_{total} decreases and therefore relative catalyst degradation exceeds 100%.

Cell	$\delta_{j-E,\text{exp}}/\mu \text{ V h}^{-1}$	$\delta_{j-E,\text{mod}}/\mu \text{ V h}^{-1}$	Due to catalyst/%	Due to resistance (R_{total})/%
A	8.2	10.2	177.2 ^{a)}	−77.2 ^{a)}
B	114.3	123.3	17.3	82.7
C	92.3	93.8	27.3	72.7
D	21.1	22.7	66.8	33.2
E	66.4	69.0	28.7	71.3

form within the anode catalyst layer, then migrate through the anode and membrane via the proton conducting ionomer and finally settle in the cathode. The Ti ions present in the anode as contaminants were suggested to cause the reduction of the anodic j_0 , either by exchanging protons with the ionomer [29,30] or changing the ratio of Ir:Ti from a more active to a less active one.

To investigate if poisoning with Ti species is present in all five CCMs investigated, transversal line scans were carried out on embedded cross sections of the CCMs. Line scans were performed in the active (aged) and passive (fresh) region of the cross section, as shown at the top of Fig. 5.

All CCMs have in common that iridium (anode) and platinum (cathode) are only found in their respective electrodes. Their distribution is equal in the passive and the active regions. However, the distribution of titanium clearly changed during the test and this is informative. A significant amount of titanium is present in the active (aged) cathodes, while none is present in the passive (fresh)

cathodes. It can be concluded that Ti poisoning occurs independent of the current density profile (Fig. 5). This corresponds well with the decrease in j_0 in Fig. 4a) that also happened for all cells and is most pronounced during the first 168 h.

Poisoning even occurs for profile A. This seems like a contradiction at first since it showed stable cell and half-cell performance. However, Fig. 4a) indicates that the drop in j_0 of cell A is completed during the first 168 h during which no half-cell data could be recorded. After 210 h, all quantities (half-cell potentials, j_0 , R_{total}) show constant values over time, indicating that migration of Ti species mostly occurs at the beginning of the experiment.

Additionally, the size of the cathode platinum particles was analyzed, as we discovered that they grow in the CCM of cell B. An example of the cathode catalyst morphology before and after the long-term test is given for cell C in Fig. 6. It is obvious that the particles grew in size and Table 5 shows that they did so for all CCMs in the aged regions. Pt particle size increased from 4.3 nm

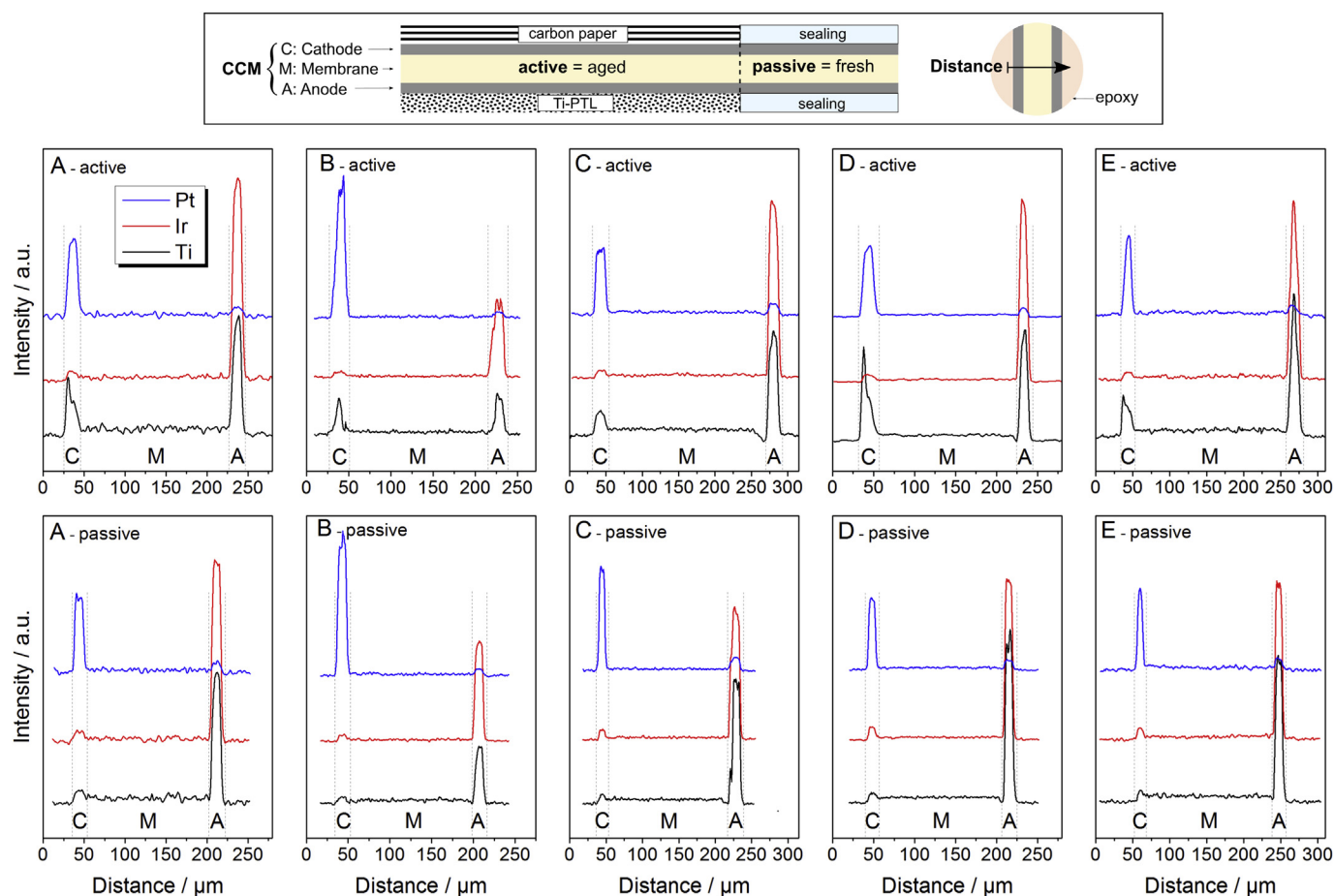


Fig. 5. EDX linescans across the CCMs of cells A–E after the long-term test. Electrolysis and ageing occurred in the active region. The elemental distribution in the passive region is representative of the fresh CCM. Scans were performed at CCMs that were embedded in epoxy resin. It is obvious that titanium is present in the cathode as a contaminant. Figures B: modified from Ref. [12] with permissions from Elsevier.

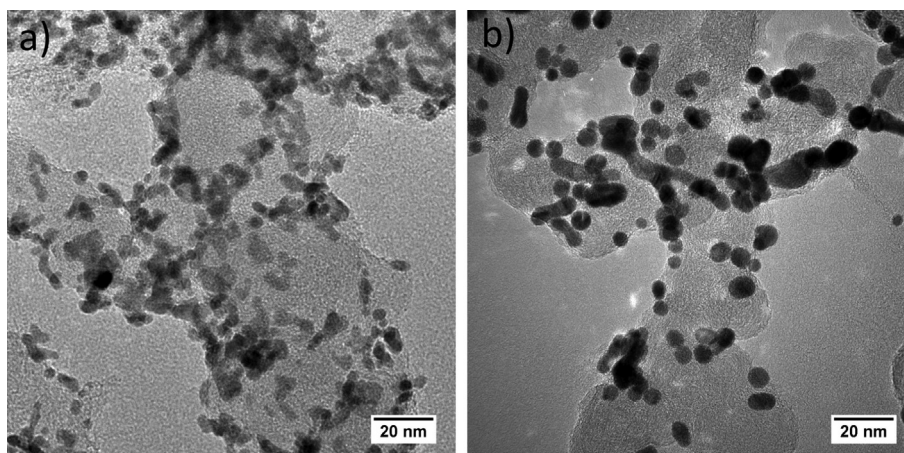


Fig. 6. TEM images of cathode catalyst of cell C. a) Fresh CCM; b) Aged CCM after 1000 h of operation. Pt particle growth and agglomeration can be observed.

Table 5

Size of catalyst particles, investigated by two methods. The crystallite size was determined through XRD and particle size (at least 400 particles) through TEM. Standard deviations of the particle size distributions are in parentheses.

	Anode XRD $d_{\text{IrO}_2}/\text{nm}$	Cathode XRD d_{Pt}/nm	Cathode TEM d_{Pt}/nm
Fresh	3.9 ± 0.2	3.5 ± 0.2	4.3 ± 0.9
Cell A	3.9 ± 0.2	7.3 ± 0.2	6.9 ± 1.6
Cell B	3.9 ± 0.2	7.8 ± 0.2	7.7 ± 2.4
Cell C	3.9 ± 0.2	7.8 ± 0.2	6.5 ± 1.6
Cell D	3.9 ± 0.2	8.1 ± 0.2	6.7 ± 1.5
Cell E	3.9 ± 0.2	14.8 ± 0.2	8.8 ± 2.5

(fresh) to 8.8 nm (aged), even for the cell operated at 1 A cm^{-2} that did not show an increase in cell voltage. Apart from cell E, the Pt particle size is in good agreement with the Pt crystal size found for the scratched off catalyst (Table 5), indicating that the imaged particles are crystalline. With the cathode of cell E, the measured crystallite size (14.8 nm) is considerably larger than the imaged particle size (8.8 nm). This difference might be due to a nonuniform particle growth, a hypothesis supported by the high standard deviation of the particle size distribution of 2.5 nm for cell E. It is the cell with the highest number of set-point changes and also exhibits the largest growth of Pt particles. This result corresponds to the decrease in ECSA that was reported for an operation with frequent current interruptions [31]. It also shows that frequent current interruptions increase cathode degradation and should therefore be avoided. The anode catalyst crystallite size remained constant throughout the test period.

4. Conclusion

Five identical single cells, each with a different current density profile, were operated in order to compare the effect of constant and dynamic power operation on performance durability. Three distinctive outcomes were identified for the cells:

- Medium current density (1 A cm^{-2} at 1.7 V): No degradation is seen. The observed stable cell voltage results from the superposition of an improved ohmic resistance and a degraded exchange current density j_0 , both effects being completed within the first 168 h of operation. Since the resistance of the titanium porous transport layer (Ti-PTL) showed no sign of degradation, the potential use of expensive anti-corrosion coatings on the Ti-PTL can be avoided for this operation mode.

- High constant current density (2 A cm^{-2} at about 1.9 V): Highest degree of degradation observed. Increasing ohmic losses at the Ti-PTL and decreased j_0 due to electrode poisoning. Reversible degradation phenomena were identified as the cause of the voltage increase.
- Dynamic current density: Degradation is reduced compared to constant 2 A cm^{-2} , when the current is reduced periodically. It was found that: i) Reducing, but not stopping the current periodically, improves durability by reducing the buildup of ohmic losses; ii) Periodically interrupting the current completely improves the durability even further. Both the time-based and the produced hydrogen-based degradation rates are reduced significantly. However, more or shorter interruptions appear to enhance cathode degradation, and therefore less or longer pauses should be undertaken.

In this paper we present that, when assembled with pure Ti-PTL, the performance durability of our PEM water electrolysis cells can be controlled by using the appropriate power profiles. Therefore, dynamic current density profiles, as they may occur when coupled to renewable energy sources, have the potential to affect positively the durability of PEM water electrolysis.

Acknowledgement

The authors are grateful for support from the Federal Ministry for Economic Affairs and Energy (BMWi) under Grant 03ESP106A. We also thank Jan Byrknes and Dr. Christian Eickes from Greenerity GmbH for the inspiring discussions and CCM samples. Furthermore, we thank Daniel Holtz (components fabrication), Michelle Poestges and Fabian Tigges (cell assembly and testing) and Katja Klafki (sample preparation).

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2016.11.118>.

References

- [1] G. Li, H. Yu, X. Wang, S. Sun, Y. Li, Z. Shao, B. Yi, Highly effective $\text{Ir}(\text{x})\text{Sn}(1-\text{x})\text{O}_2$ electrocatalysts for oxygen evolution reaction in the solid polymer electrolyte water electrolyser, *Phys. Chem. Chem. Phys.* 15 (8) (2013) 2858–2866.
- [2] H. Su, V. Linkov, B.J. Bladergroen, Membrane electrode assemblies with low noble metal loadings for hydrogen production from solid polymer electrolyte water electrolysis, *Int. J. Hydrogen Energy* 38 (23) (2013) 9601–9608.
- [3] M.K. Debe, S.M. Hendricks, G.D. Vernstrom, M. Meyers, M. Brostrom,

- M. Stephens, Q. Chan, J. Willey, M. Hamden, C.K. Mittelsteadt, C.B. Capuano, K.E. Ayers, E.B. Anderson, Initial performance and durability of ultra-low loaded NSTF electrodes for PEM electrolyzers, *J. Electrochem. Soc.* 159 (6) (2012) K165.
- [4] H. Ma, C. Liu, J. Liao, Y. Su, X. Xue, W. Xing, Study of ruthenium oxide catalyst for electrocatalytic performance in oxygen evolution, *J. Mol. Catal. A Chem.* 247 (1–2) (2006) 7–13.
- [5] C. Rozain, E. Mayousse, N. Guillet, P. Millet, Influence of iridium oxide loadings on the performance of PEM water electrolysis cells: part II – advanced oxygen electrodes, *Appl. Catal. B* 182 (2016) 123–131.
- [6] R.E. Clarke, S. Giddey, S. Badwal, Stand-alone PEM water electrolysis system for fail safe operation with a renewable energy source, *Int. J. Hydrogen Energy* 35 (3) (2010) 928–935.
- [7] R.E. Clarke, S. Giddey, F.T. Ciacchi, S. Badwal, B. Paul, J. Andrews, Direct coupling of an electrolyser to a solar PV system for generating hydrogen, *Int. J. Hydrogen Energy* 34 (6) (2009) 2531–2542.
- [8] T. Audichon, E. Mayousse, T.W. Napporn, C. Morais, C. Comminges, K.B. Kokoh, Elaboration and characterization of ruthenium nano-oxides for the oxygen evolution reaction in a Proton Exchange Membrane Water Electrolyzer supplied by a solar profile, *Electrochim. Acta* 132 (2014) 284–291.
- [9] K.E. Ayers, J.N. Renner, N. Danilovic, J.X. Wang, Y. Zhang, R. Maric, H. Yu, Pathways to ultra-low platinum group metal catalyst loading in proton exchange membrane electrolyzers, *Catal. Today* 262 (2016) 121–132.
- [10] S. Sun, Z. Shao, H. Yu, G. Li, B. Yi, Investigations on degradation of the long-term proton exchange membrane water electrolysis stack, *J. Power Sources* 267 (2014) 515–520.
- [11] H. Su, B.J. Bladergroen, V. Linkov, S. Pasupathi, S. Ji, Study of catalyst sprayed membrane under irradiation method to prepare high performance membrane electrode assemblies for solid polymer electrolyte water electrolysis, *Int. J. Hydrogen Energy* 36 (23) (2011) 15081–15088.
- [12] C. Rakousky, U. Reimer, K. Wippermann, M. Carmo, W. Lueke, D. Stolten, An analysis of degradation phenomena in polymer electrolyte membrane water electrolysis, *J. Power Sources* 326 (2016) 120–128.
- [13] G. Wei, Y. Wang, C. Huang, Q. Gao, Z. Wang, L. Xu, The stability of MEA in SPE water electrolysis for hydrogen production, *Int. J. Hydrogen Energy* 35 (9) (2010) 3951–3957.
- [14] X. Wang, L. Zhang, G. Li, G. Zhang, Z.-G. Shao, B. Yi, The influence of Ferric ion contamination on the solid polymer electrolyte water electrolysis performance, *Electrochim. Acta* 158 (2015) 253–257.
- [15] P. Millet, R. Ngameni, S.A. Grigoriev, N. Mbemba, F. Brisset, A. Ranjbari, C. Etiévant, PEM water electrolyzers: from electrocatalysis to stack development, *Int. J. Hydrogen Energy* 35 (10) (2010) 5043–5052.
- [16] ITM POWER, The development of a PEM electrolyzer AST (11.12.2014), URL www.electrohypem.eu/data/ITM_Nick-van-Dijk.pdf.
- [17] J. Giner, A practical reference electrode, *J. Electrochem. Soc.* 111 (3) (1964) 376.
- [18] J.H. Ohs, U. Sauter, S. Maass, D. Stolten, The Effect of the reference electrode position on the measurement of half cell polarization in proton-exchange membrane fuel cells, *J. Electrochem. Soc.* 159 (7) (2012) F181–F186.
- [19] Z. Siroma, R. Kakitsubo, N. Fujiwara, T. Ioroi, S.-I. Yamazaki, K. Yasuda, Compact dynamic hydrogen electrode unit as a reference electrode for PEMFCs, *J. Power Sources* 156 (2) (2006) 284–287.
- [20] M. Carmo, T. Roepke, F. Scheiba, C. Roth, S. Moeller, H. Fuess, J.G. Poco, M. Linardi, The use of a dynamic hydrogen electrode as an electrochemical tool to evaluate plasma activated carbon as electrocatalyst support for direct methanol fuel cell, *Mater. Res. Bull.* 44 (1) (2009) 51–56.
- [21] M. Pourbaix, *Atlas of Electrochemical Equilibria in Aqueous Solutions*, Nat'L Assoc. Of Corrosion, 1974.
- [22] S. Cherevko, T. Reier, A.R. Zeradjanin, Z. Pawolek, P. Strasser, K.J. Mayrhofer, Stability of nanostructured iridium oxide electrocatalysts during oxygen evolution reaction in acidic environment, *Electrochem. Commun.* 48 (2014) 81–85.
- [23] L. Zhang, X. Jie, Z.-G. Shao, X. Wang, B. Yi, The dynamic-state effects of sodium ion contamination on the solid polymer electrolyte water electrolysis, *J. Power Sources* 241 (2013) 341–348.
- [24] S. Adler, Reference electrode placement and seals in electrochemical oxygen generators, *Solid State Ion.* 134 (1–2) (2000) 35–42.
- [25] S.B. Adler, Reference electrode placement in thin solid electrolytes, *J. Electrochem. Soc.* 149 (5) (2002) E166.
- [26] W. He, T. van Nguyen, Edge effects on reference electrode measurements in PEM fuel cells, *J. Electrochem. Soc.* 151 (2) (2004) A185.
- [27] U. Diebold, The surface science of titanium dioxide, *Surf. Sci. Rep.* 48 (5–8) (2003) 53–229.
- [28] R.P. O'Hayre, S.-W. Cha, W.G. Colella, F.B. Prinz, *Fuel Cell Fundamentals*, second ed., Wiley, New York, 2009.
- [29] B.L. Kienitz, H. Baskaran, T.A. Zawodzinski, Modeling the steady-state effects of cationic contamination on polymer electrolyte membranes, *Electrochim. Acta* 54 (6) (2009) 1671–1679.
- [30] B. Kienitz, H. Baskaran, T. Zawodzinski, B. Pivovar, A Half Cell Model to Study Performance Degradation of a PEMFC due to Cationic Contamination, in: 212th ECS Meeting, October 7–October 12, 2007, pp. 777–788.
- [31] E. Brightman, J. Dodwell, N. van Dijk, G. Hinds, In situ characterisation of PEM water electrolyzers using a novel reference electrode, *Electrochem. Commun.* 52 (2015) 1–4.